

Econometrics Foundation

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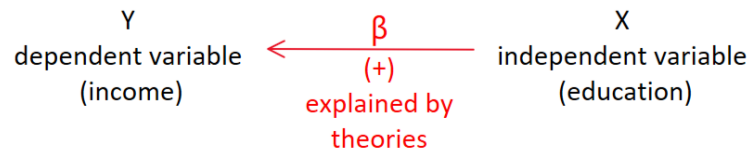
1 What problem does econometrics solve?

"Econometrics is used in **academic research** for **quantitative analysis** of the **relationships(correlation) and causality** between **economic variables** or a **pair of factor and outcome** that can be **expressed numerically**."

e.g. Suppose we want to study the **impact** of **education level** on individual **income**.

1. **Data collection:** We collect data on the number of years of education (such as years of schooling) and their annual income for a group of people.
2. **Model construction:** Using econometric regression analysis methods, we build a model to describe the relationship between years of education and income.
3. **Result analysis:** Through model estimation (estimates the value and significance of the correlation coefficient between them), we may find that for each additional year of education, the average income increases by a certain amount, i.e. there is a statistically significant positive parameter.
4. **Interpretation:** This indicates that education has a positive impact on increasing income, which verifies the human capital theory¹ in economics.

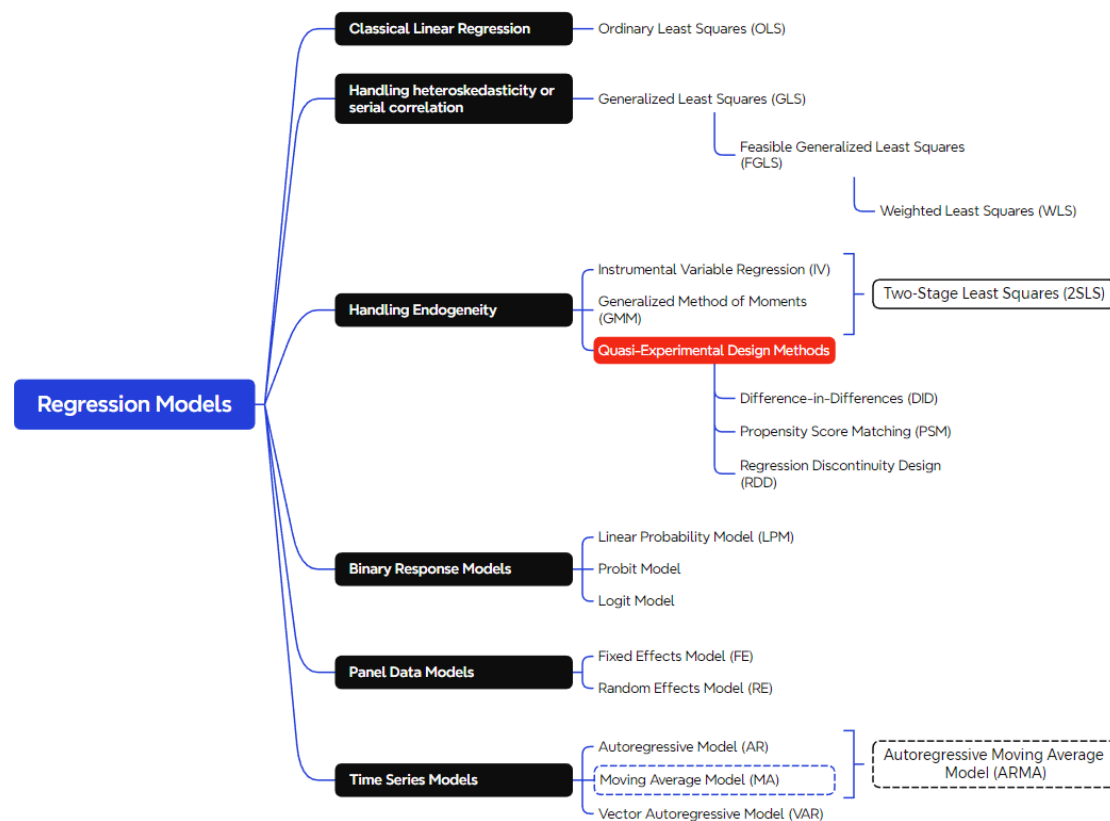
¹ **Human Capital Theory:** Education increases skills, making individuals more productive, leading to higher earnings.



Thus, under this **econometric research paradigm**, we primarily use **statistical methods** to **estimate parameters** when studying the **correlation and causality research question**, and then use **appropriate economic theories** to explain these parameters.

2 How to estimate the parameters?

We use a series of **regression models** or **regression analysis frameworks**:



This is easy to generate results from statistical analysis software (SPSS, Stata, Python, R, MATLAB etc.), if we have learn the code syntax. However, to get plausible, convincing, realistic, true (**unbiased** & **precise(or efficient)**) results require a rigorous research **design, implementation** and **analysis** process.

In the paradigm of econometrics, there is a well-established set of assumptions that, if met, can help us:

Design proper and rigorous **research**, e.g. experimental and non-experimental design;

Collect and process more **reliable data**, e.g. Selection and measurement of variables, sample design;

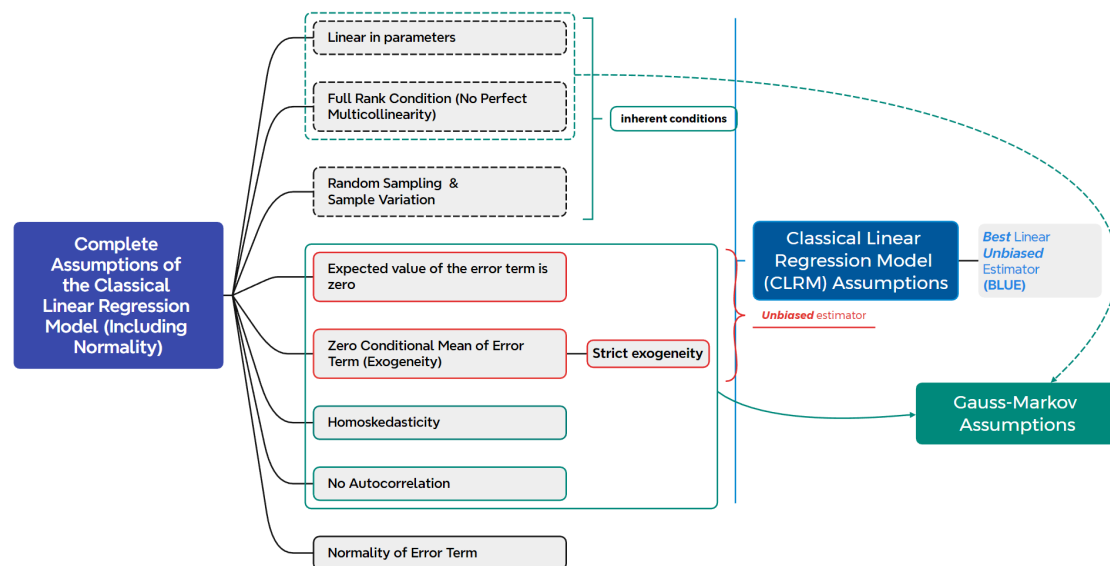
Select and apply the most appropriate **models and methods**;

Provide more **accurate and meaningful interpretations** of the **results**, e.g. , causal inference, reliability of results.

3 How to ensure the parameter estimators of the regression model have good statistical properties and reliable statistical inference?

Assumption Framework :

These assumptions ensure that the OLS estimator is unbiased, consistent, and efficient in the context of a simple linear regression model. They provide the basis for hypothesis testing and confidence interval construction.



1. The Most Stringent Assumption Framework: Complete Assumptions of the Classical Linear Regression Model (Including Normality)

- Under these assumptions, the **Ordinary Least Squares (OLS) estimator is the *Best(efficient)* Linear *Unbiased* Estimator (BLUE)** and has the smallest variance among all unbiased linear estimators.
- The parameter estimates have exact **normal distributions**, allowing for precise statistical inference (e.g., t-tests and F-tests) even in **small samples**.

2. Classical Linear Regression Model (CLRM) Assumptions

This framework **relaxes** the normality assumption of the error term compared to the most stringent framework.

- **Because** the exact distribution of the parameter estimates is **unknown**, but by the **Central Limit Theorem²**, the estimates are **approximately normally distributed** in **large samples**.
- **Statistical inference** relies on **asymptotic theory**, and may not be **precise** in small samples.

3. Gauss-Markov Assumptions

² **Central Limit Theorem (CLT)** states that, for a sufficiently large sample size, the distribution of the sample mean approaches a **normal distribution**, regardless of the original population distribution. It allows us to make statistical inferences about sample means using normal distribution, making hypothesis testing and confidence interval construction simpler and more possible.

This framework **further relaxes** the assumptions by dropping the random sampling assumption.

- **Random sampling is not required**; observations can be **non-identically distributed**. Similarly, **Linear in parameters** and **no perfect multicollinearity** are also regarded as inherently satisfied conditions in practice as they are actually the basic structure or mathematical conditions of regression models, which are often naturally required when building regression models.
- Statistical inference still relies on **large sample properties**³
- Under these assumptions, the **OLS estimator is still BLUE**.

Assumptions list:

1. The structure and parameters of the linear regression model

1) **Linear Model in Parameters**: The model is linear in its parameters, meaning the dependent variable is a linear combination of the independent variables plus an error term: $Y = \alpha + \beta_1 X + \varepsilon$

2) **Full Rank Condition (No Perfect Multicollinearity)**: The **matrix** of independent variables has **full rank**⁴; the columns are **linearly independent**, indicating that in a set of regressors x 's matrix X , one regressors are not exactly the combinations of others.

E.g. we cannot put all of variable gender, male, female in the model at the same time.

³ **Large Sample Properties** refer to the **behavior of estimators** as the **sample size grows**. Key properties include:

1. **Consistency**: The estimator **converges to the true parameter value** as the sample size increases.
2. **Asymptotic Normality**: The **estimator's distribution approaches a normal distribution** as the sample size grows, allowing the use of normal distribution for inference.

It is also fundamental for justifying statistical inference in large samples, enabling reliable estimation and hypothesis testing.

⁴ Full rank: the rank of matrix is the number of linearly independent rows or columns in the matrix. Full rank occurs when the rank is equal to the number of rows or columns.

Meanwhile, in OLS calculation, especially using **matrix** to deal with multiple linear regression, it requires matrix to be full rank and invertible for derivations and proofs.

2. Sampling process of data

3) **Random Sampling and Sample Variation:** The sample is randomly drawn from the population; **observations** are **independently and identically distributed**⁵ (i.i.d), and some **variations** in the regressors **y** & **x**.

The OLS estimator is

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{Cov(x_i, y_i)}{Var(x_i)}$$

We use R^2 indicates the proportion of the variance in **y** that can be explained by the linear combination of **x** variables.

3. Characteristics of the error term

4) Unconditional: **Expected value of the error term is zero:** Given the independent variables, the expected value of the error term is zero: $E(\varepsilon_i) = 0$, $i = 1, \dots, N$

5) Conditional: **Zero Conditional Mean of Error Term (Strict Exogeneity):** Given the independent variables, the expected value of the error term is zero: $E(\varepsilon_i | X_i) = 0$

*Satisfying these 2 assumptions, the estimator β will be **unbiased**. Prove in

⁵ **Independently and Identically Distributed (i.i.d.)** refers to a set of random variables that are both:

1. **Independent:** Each variable is not influenced by any of the others; their values are unrelated.
2. **Identically Distributed:** All variables follow the same probability distribution, with the same statistical properties (e.g., mean, variance).

It's crucial for the **Central Limit Theorem**, which allows sample means to approximate a **normal distribution** when the sample is sufficiently large

OLS, i.e., minimizing the **difference** between the **observed value** y and its **linear approximation** $X\hat{\beta}$, i.e., $y - \hat{\beta}X$. As $y = \beta X + \varepsilon$, the optimized $\hat{\beta}$ is the $\hat{\beta}^*$ that mostly close to the real β .

$$S(\hat{\beta}) \equiv (y - \hat{\beta}X)'(y - \hat{\beta}X)$$

$$\min (y - \hat{\beta}X)'(y - \hat{\beta}X)$$

First order conditions = 0

$$\hat{\beta}^* = b = (X'X)^{-1}(X'y) = (X'X)^{-1}(X\beta + \varepsilon)$$

$$b = \beta + (X'X)^{-1}X'\varepsilon$$

Taking expectations on both sides (conditional upon X)

$$E(b|X) = E(\beta + (X'X)^{-1}X'\varepsilon|X)$$

$$= E(\beta) + (X'X)^{-1}X'E(\varepsilon|X)$$

$$= \beta + 0$$

6) **Homoskedasticity**: Given the independent variables, the variance of the error term is constant: $Var(\varepsilon_i|x_i) = \sigma^2$ or $\sigma^2 I_N$, $i=1, \dots, N$

7) **No Autocorrelation**: The error terms are uncorrelated across observations:

$$Cov(\mu_i, \mu_j) = 0, i, j=1, \dots, N, i \neq j$$

Combination of them is **Spherical variance**:

- $Var(\varepsilon) = E(\varepsilon\varepsilon')$ is the $N \times N$ variance-covariance matrix of the errors i.e.

$$Var(\varepsilon) = \begin{pmatrix} \varepsilon_1^2 & \varepsilon_1\varepsilon_2 & \dots & \varepsilon_1\varepsilon_N \\ \varepsilon_2\varepsilon_1 & \varepsilon_2^2 & \dots & \varepsilon_2\varepsilon_N \\ \vdots & \vdots & \ddots & \vdots \\ \varepsilon_N\varepsilon_1 & \varepsilon_N\varepsilon_2 & \dots & \varepsilon_N^2 \end{pmatrix}$$

Covariance = 0
(no autocorrelation)

- Taken together, assumptions (A3) and (A4) imply that $Var(\varepsilon) = \sigma^2 I_N$, i.e.

$$Var(\varepsilon) = \begin{pmatrix} \sigma^2 & 0 & \dots & 0 \\ 0 & \sigma^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma^2 \end{pmatrix}$$

Variance = σ
(Homoskedasticity)

$$\begin{aligned}
Var(b|X) &= E\{(b - \beta)(b - \beta)'\} = E\{(X'X)^{-1}X' \varepsilon \varepsilon' X (X'X)^{-1}\} \\
&= E\left\{(X'X)^{-1}X' \sigma^2 I_N X (X'X)^{-1}\right\} \\
&= \sigma^2 I_N (X'X)^{-1}
\end{aligned}$$

8) **Normality of Error Term:** Given the independent variables X , the error term μ or error terms are jointly normally distributed: $\mu \sim Normal(0, \sigma^2)$ or $\mu \sim Normal(0, \sigma^2 I_N)$

Therefore, if assumption 4),5),6),7) all hold true, and we also assume Normality of Error Term hold, we will know the **distribution of estimator b** , with mean of $E(b|X) = \beta + 0$ and variance $Var(b|X) = \sigma^2 I_N (X'X)^{-1}$, i.e., $b \sim Normal(\beta, \sigma^2 I_N (X'X)^{-1})$ or $b_k \sim Normal(\beta, \sigma^2 c_{kk})$ where c_{kk} is (k,k) element in $(X'X)^{-1}$.

Finally, we will use **test statistics** to get **inference** of estimate value and **significance**.

$$z_k = \frac{b_k - \beta_k^0}{\sqrt{\sigma^2 c_{kk}}} \sim Normal(0,1)$$

Where z_k is **test statistic**, β_k^0 denotes hypothesis value (usually 0)

e.g. **null hypothesis: $b_k = \beta_k^0 = 0$** , i.e., explanatory variables x have no significant effect on dependent variables y ;

and most commonly used test statistic **t statistic** $z_k \rightarrow t_k$, as we usually do not know the σ^2 in practice, and use its estimates $s^2 = \frac{1}{N-K} \sum_{i=1}^N e_i^2$ and get

standard error $\sqrt{s^2 c_{kk}} = se(b_k)$, where e is the observed **residuals**.

$$t_k = \frac{b_k - \beta_k^0}{se(b_k)} \sim t(N - K)$$

Where $N-K$ is the **degree of freedom**, when it is larger the t distribution more approximate Normal distribution. Usually, when sample N is large enough we

can get inference by combining t-statistic t_k and critical value (e.g. 1.96 at significant level $\alpha = 0.05$ with 95% confidence interval).

$$\text{confidence interval: } [b_k - 1.96 * se(b_k), b_k + 1.96 * se(b_k)]$$

When the hypothesis value is not in the confidence interval, we can conclude that at the 95% confidence level, we have sufficient evidence to **reject the null hypothesis**. This means that the parameter β is most likely not 0, indicating that the explanatory variable x has a significant effect on the dependent variable y.

